

DESIGNING A THREE PHASE, CORE TYPE TRANSFORMER

Course Title : Electrical Machine Design

Course Code: EE3220

SUBMITTED TO:

Dr. Md. Habibullah

Professor,

Department of Electrical and Electronic Engineering. Khulna University of Engineering & Technology (KUET)

Dr. Md. Nur Kutubul Alam

Associate Professor,

Department of Electrical and Electronic Engineering. Khulna University of Engineering & Technology (KUET)

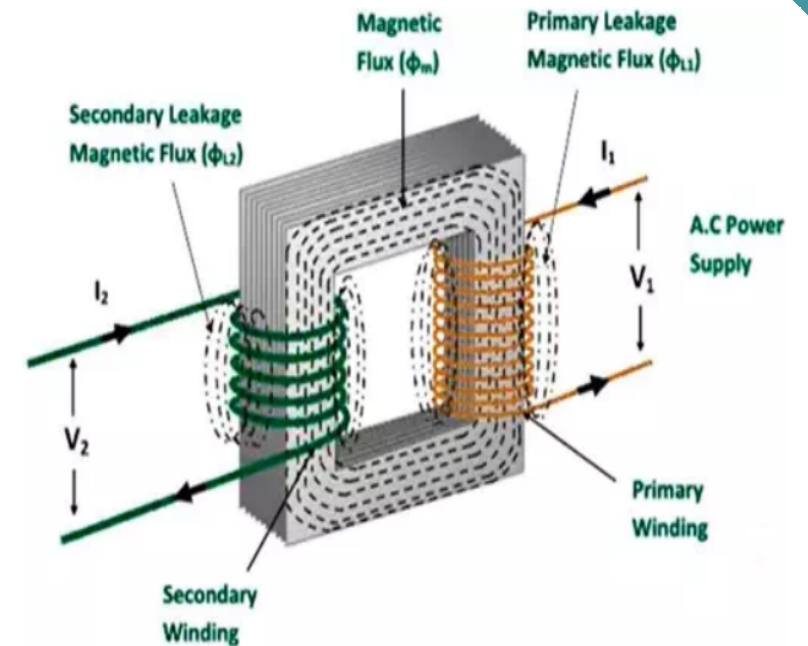
SUBMITTED BY:

SL. No.	Name	Roll
1.	MD.MAHMUD-UN-NABI SAZID	2003116
2.	MD.SAKIBUL HOQUE	2003117
3.	MD. TOUKIR AHMMED	2003118
4.	MD. ARAFAT BEN SHARAFAT	2003119
5.	ABDULLAH AL SOYKOT	2003120
6.	DARPON CHAKMA	2003121

Introduction:

A transformer is a stationary electrical device that transfers alternating current (AC) power from one circuit to another while maintaining a constant frequency, without any direct electrical connection. It can adjust the voltage level as needed, either increasing or decreasing it.

This process works by inducing an electromotive force (EMF) in the secondary circuit due to the changing magnetic flux in the secondary winding.

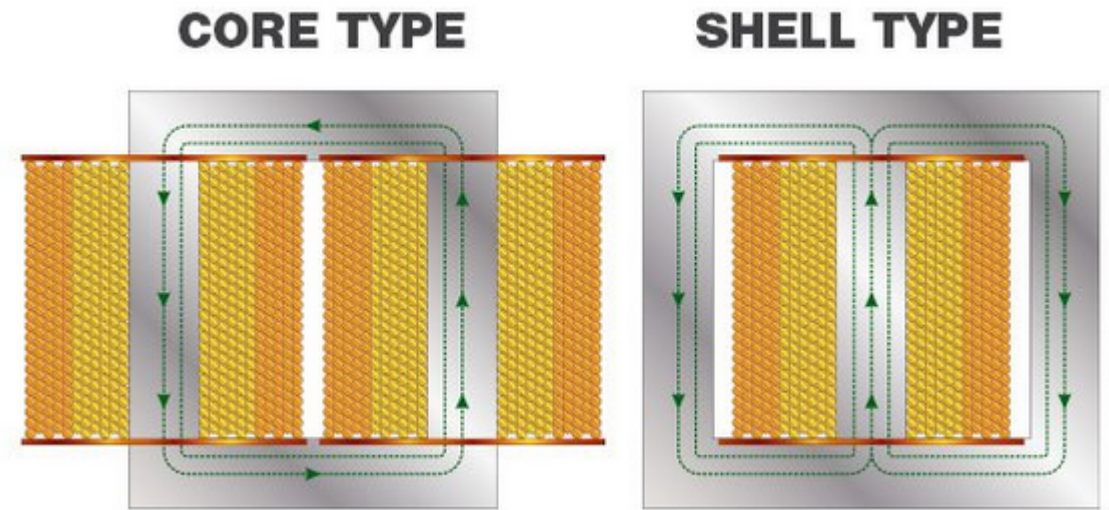


Types of Transformer:

A transformer is essentially composed of a **magnetic core** and two windings called **primary and secondary winding**.

On the basis of construction, transformers can be classified into two types:

- (i) Core type transformer
- (ii) Shell type transformer



Core type & Shell type Transformer:

Core type transformer:

1. Core is surrounded by winding.
2. Concentric winding is used.
3. Width of all limbs is equal.

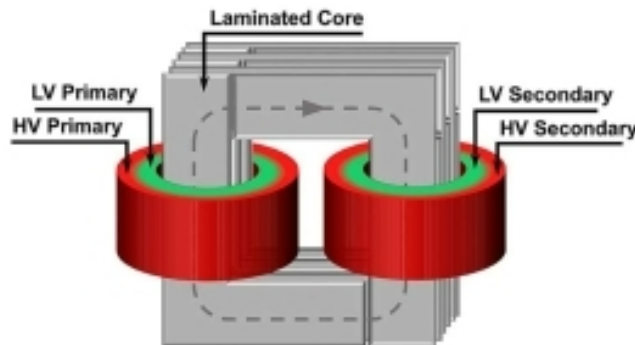


Fig. Core Type Transformer

Shell type transformer:

- 1) winding is surrounded by core.
- 2) Sandwich winding is done.
- 3) Width of central limb is greater than that of side limbs.

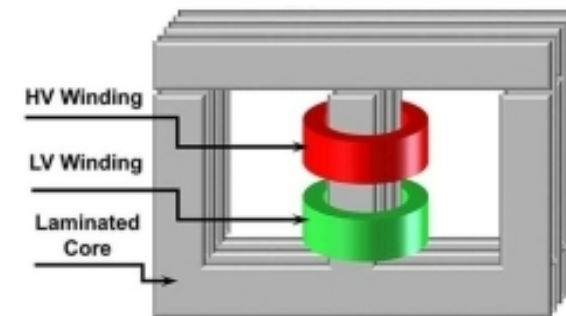
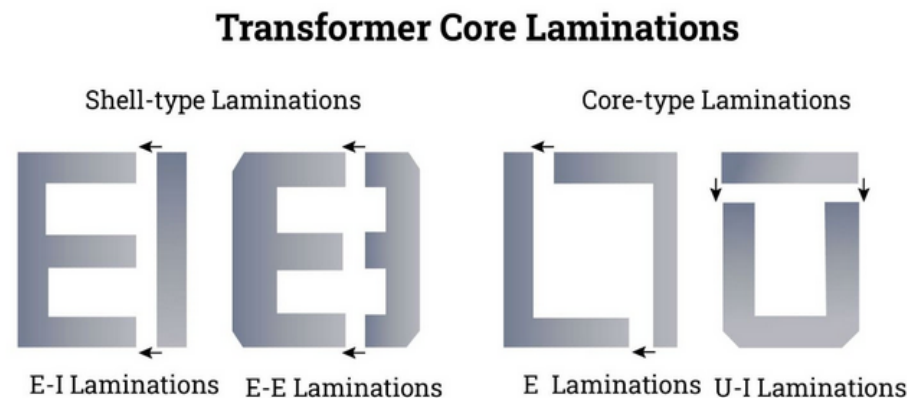


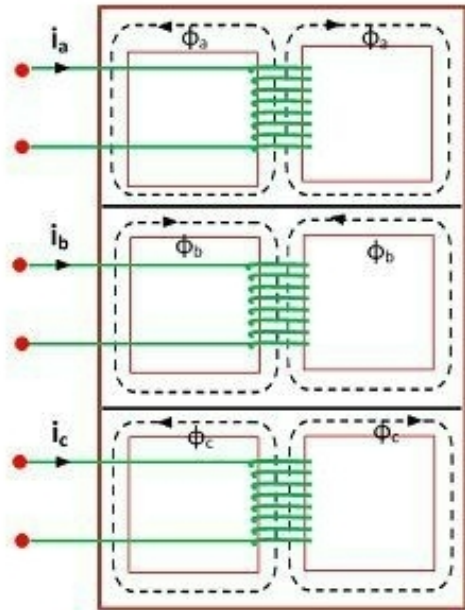
Fig. Shell Type Transformer

Types of Laminated Core:

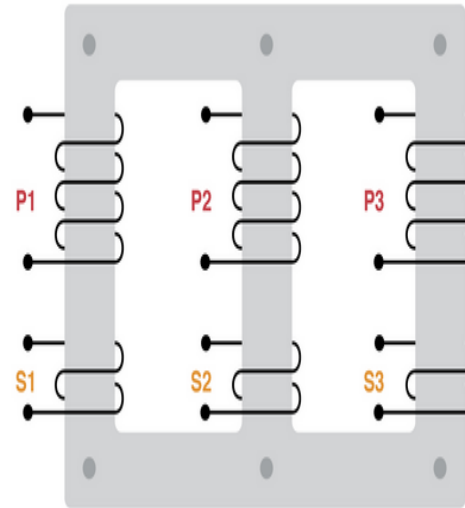
- The core is constructed of numerous thin strips of grain-oriented silicone steel called laminations (96-97% Fe, 3-4% Si).
- These laminations are available in various shapes such as- E, I, L, O, C.
- Their height and width size varies depending upon the transformer-type.



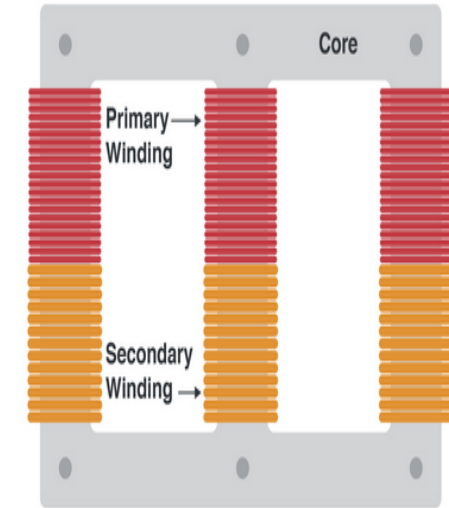
Three Phase Winding:



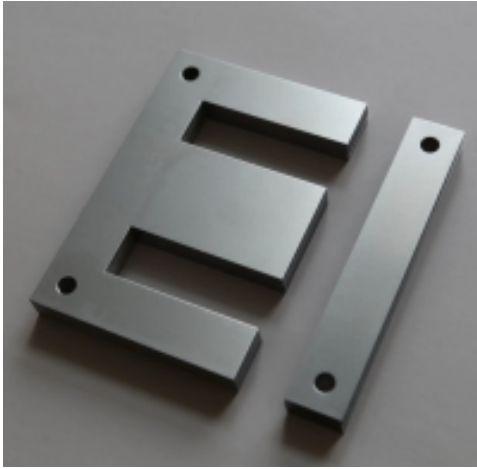
3-Phase Core Type Transformer



3-Phase Core Type Transformer



Required Components:



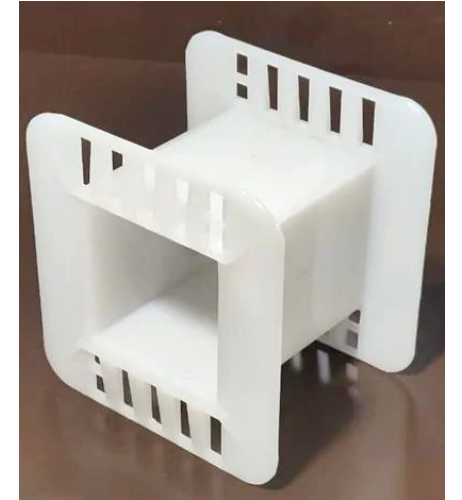
Laminations



Insulation Tape



wire



Bobbin

Required Apparatus:

SL.No.	Name of the Apparatus	Quantity
1.	Core laminations (E,I)	81
2.	Paper insulation	As required
3.	Bobbin	03
4.	Copper wire	(640+150) gm
5.	Insulation Tape	As required
6.	Nuts and bolts	4

DESIGN SPECIFICATION:

- Three-phase Core-type transformer
- Total apparent power : 165 VA
- Intermittent operation
- Primary voltage : 230 V Y - Y
- Secondary voltage : 24 V Y
- Frequency : 50 Hz
- Cooling : air

Core Design:

Let, utilization coefficient, $K_u = 1.9 \times 10^{-2}$ [for core type three-phase],

$$\text{Flux, } \Phi = K_u \sqrt{(KVA/f)} = 1.9 \times 10^{-2} \sqrt{(0.165/50)} = 1.0915 \times 10^{-3} \text{ Wb}$$

Flux Density, $B = 1 \text{ Wb/m}$ [for intermittent operation]

$$\begin{aligned} \text{Area of the net iron section, } A_{ir} &= \Phi/B = 1.0915 \times 10^{-3} / 1 \text{ m}^2 \\ &= 1.0915 \times 10^{-3} \text{ m}^2 \\ &= 1092 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{The column width (approximately), } C &= \sqrt[4]{VA/3} = \sqrt[4]{165/3} \\ &= 2.72 \text{ cm} = 27.2 \text{ mm} \approx 30 \text{ mm} \end{aligned}$$

A commercial lamination of external size 150 x 150, column size 30 x 90 and window size 30 x 90, having 0.5 mm thickness including the paint insulation has been selected.

Core Design (cont'd):

The net thickness of the iron package, $L_p = A_{ir} / C = (1092 / 30) \text{ mm}$
 $= 36.4 \text{ mm}$

Let, packing coefficient, $K_s = 0.9$

∴ Gross thickness is $L_{po} = L_p / K_s = (36.2 / 0.9) \text{ mm}$
 $= 40.44 \text{ mm}$

∴ No. of laminations $= 40.44 / 0.5 = 80.88 \approx 81$ laminations

To mount the core, the laminations are inserted into a 30 x 50 nylon reel, one for each single phase.

Core Design (cont'd):

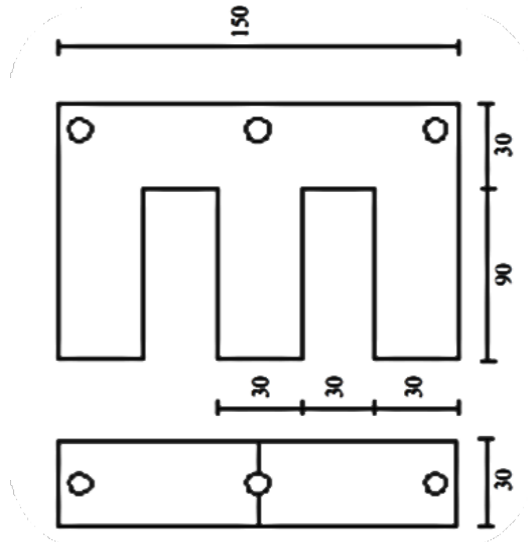


Fig. Core Size

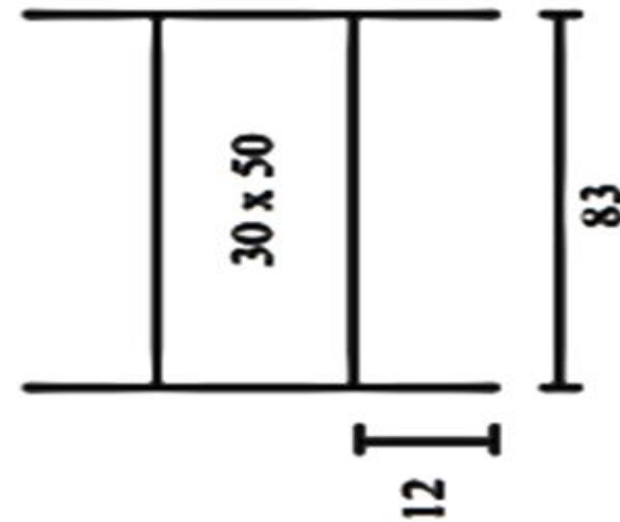


Fig. Reels for 3-phase Winding

Winding Design:

Secondary phase voltage, $V_{2ph} = 24/\sqrt{3} \text{ V} = 13.856 \text{ V}$

The voltage per turn, $e = 4.44 f \Phi = 4.44 \times 50 \times 1.0915 \times 10^{-3} = 0.2423 \text{ V}$

The no. of primary turns/phase, $N_1 = V_{1ph} / e = 132.79 / 0.2423$

$= 548.039$

$\approx 549 \text{ turns}$

Let, voltage drop from no-load to load condition = 3.5%

The no. of secondary turns/phase, $N_2 = [V_{2ph} + \Delta] / e$

$= [13.856 + 0.48496] / 0.2423$

$= 59.18$

$\approx 60 \text{ turns}$

Winding Design (cont'd):

Let, efficiency, $\eta = 91\%$ and current density, $\delta = 3.5 \text{ A/mm}^2$

Primary current/phase,

$$I_{1ph} = S/[V_{1ph} * \eta] = \frac{165/3}{132.79 \times 0.91} = 0.455 \text{ A}$$

Copper section, $A_{cu1} = (I_{1ph} / \delta) = 0.13 \text{ mm}^2$

$\therefore$ Diameter, $d_1 = 0.4068 \text{ mm}$

Secondary current/phase, $I_{2ph} = S/V_{2ph} = \frac{165/3}{13.856} = 3.969 \text{ A}$

Copper section, $A_{cu2} = I_{2ph} / \delta = 1.1314 \text{ mm}^2$

$\therefore$ Diameter, $d_2 = 1.2 \text{ mm}$

Winding Dimension:

Useful height of the reel, $h = 83\text{mm}$ Filling coefficient, $K_r = 1.05$

Using a wire with single enamel clad, the maximum diameters on enamel are

$$\varnothing_{1s} = 0.4568 \text{ mm}$$

$$\varnothing_{2s} = 1.25 \text{ mm}$$

So, no. of turns per layer,

$$n_1 = \frac{h}{K_r \times \varnothing_{1s}} = \frac{83}{1.05 \times 0.4568} = 173$$

$$n_2 = \frac{h}{K_r \times \varnothing_{2s}} = \frac{83}{1.05 \times 1.25} = 63$$

The number of layers,

$$\begin{aligned} S_p &= N_1/n_1 = 549/173 \\ &= 3.167 \\ &\approx 4 \end{aligned}$$

$$\begin{aligned} S_s &= N_2/n_2 = 60/63 \\ &= 0.95 \\ &\approx 1 \end{aligned}$$

Demonstration Of Winding Layer:

The radial dimension of winding,

$R = S_p \varnothing 1s + S_s \varnothing 2s + \text{paper thickness} \times (S_p + S_s - 1) + \text{lathieroid thickness} \times 2$

$$= 4 \times 0.4568 + 1 \times 1.25 + 0.1 \times (4 + 1 - 1) + 0.3 \times 2$$

$$= 4.0772 \text{ mm} < 15 \text{ mm}$$

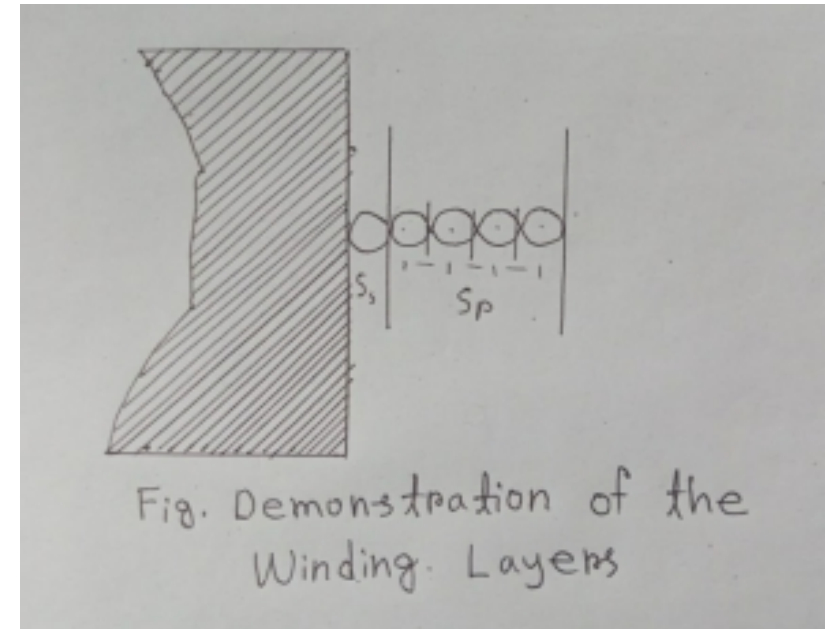


Fig : Demonstration of the Winding Layers

Limitations:

- Slight overheating
- Humming while operating
- Lower voltage than expected at secondary side

Solutions:

- Reducing air gap in the winding
- Fitting core laminations properly so that there is no gap between laminations
- Providing proper cooling arrangements







Thank you
